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PAPER_769: NGC 4676 Mice Galaxies — UQFF Dual Galaxy Merger Dynamics

Author: Daniel T. Murphy

Framework: UQFF (Universal Quantum Field Superconductive Framework)

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CP4 Class: #353 — NGC4676MiceGalaxiesDualMergerCalculator

Abstract

NGC 4676 (the “Mice Galaxies”) is a pair of colliding spiral galaxies in Coma (~ 290 Mly, $z \approx 0.022$), each $\sim 2 \times 10^{11}$ MM_{sun}, having undergone a close passage ~ 100 Myr ago. The Hubble ACS imaging (2002) reveals dramatic tidal tails, nuclear star clusters, and enhanced starburst activity triggered by the interaction. Under UQFF, the merger mass-loss function $M_{\text{merge}}(t)$, enhanced starburst magnetic field ($B \sim 10^{-4}$ T), interaction velocity ($v \sim 106$ m/s), and cosmic expansion term yield $g_{\text{Mice}} \approx 1.053 \times 10^{-1}$ m/s². The elevated magnetic field due to starburst induction is the signature UQFF distinction for colliding systems.

1. Introduction

The Mice Galaxies (NGC 4676A and NGC 4676B) represent a dual-merger system in an earlier interaction stage than the Antennae (NGC 4038/39). Both components retain distinct nuclei while exhibiting extended tidal tails reaching hundreds of kpc. N-body simulations predict they will fully merge within ~ 1 Gyr. The starburst triggered by the interaction elevates the effective magnetic field to $\sim 10^{-4}$ T (compared to isolated spirals at $\sim 10^{-6}$ T), which under UQFF produces a significantly enhanced Aether electromagnetic correction, yielding the same principal scaling as the Antennae system.

2. Master UQFF Gravity Equation

$$g_{\text{Mice}}(r, t) = (G \times M_{\text{total}})/r^2 \times (1 + H(z) \times t) \times (1 - M_{\text{merge}}) \times (1 + f_T RZ) + a_{\text{EM}} M$$

Where: - M_{total} : combined mass of both galaxies - $(1 - M_{\text{merge}})$: merger-driven mass redistribution loss factor - a_{EM} : Aether electromagnetic correction (starburst-enhanced B field)

2.1 Parameters

Parameter	Symbol	Value	Source
Total galaxy mass	M	$2 \times 10^{11} M_{\odot}$	Distance counter
		$3.978 \times 10^{11} kg$	Hubble
		$9.468 \times 10^{15} s$	Interaction radius
Merger mass fraction	M_merge	0.2638	UQFF merger func
Starburst B-field	B	10-4 T	Enhanced Starburst
Interaction velocity	v	106 m/s	UQFF Aether
$\rho_{vac}, [UA]$	—	$7.09 \times 10^{-36} J/m^3$	UQFF
f_TRZ	—	0.1	UQFF

3. Long-Form Derivation

Step 1: Base Gravitational Term

$$\begin{aligned}
 g_{grav} &= (6.6743e-11 \times 3.978e41) / (3e20)^2 \\
 &= 2.655e31 / 9e40 = 2.950e-10 m/s^2
 \end{aligned}$$

Step 2: Merger Mass Distribution Term M_merge(t)

Duringgalacticcollision, massisredistributedas :

$$\begin{aligned}
 M_{merge}(t) &= T0 \times (1 - \exp(-t/\tau_{merge})) \\
 &= 0.5 \times (1 - \exp(-3e8/4e8)) \\
 &= 0.5 \times (1 - \exp(-0.75)) \\
 &= 0.5 \times (1 - 0.4724) \\
 &= 0.5 \times 0.5276 = 0.2638 \\
 1 - M_{merge} &= 1 - 0.2638 = 0.7362
 \end{aligned}$$

Step 3: Cosmic Expansion Factor

$$\begin{aligned}
 H(z) &= H0 \times \sqrt{(\Omega_m(1+z)^3 + \Omega_\Lambda)} \\
 &= 2.268e-18 \times \sqrt{(0.3 \times (1.022)^3 + 0.7)} \\
 &= 2.268e-18 \times \sqrt{(0.3 \times 1.0677 + 0.7)} \\
 &= 2.268e-18 \times \sqrt{(1.0203)} \\
 &= 2.268e-18 \times 1.01010 = 2.291e-18 s^{-1} \\
 H(z) \times t &= 2.291e-18 \times 9.468e15 = 2.169e-2 \\
 1 + H(z) \times t &= 1.02169
 \end{aligned}$$

Step 4: Aether Electromagnetic Correction (Starburst-Enhanced)

$$\begin{aligned}
 \text{Starburstinteractionenhancesmagneticfieldto} &B = 10^{-4} T \\
 \text{Interactionvelocity} &v = 106 m/s \\
 q \times (v \times B) &= 1.602e-19 \times 1e6 \times 1e-4 = 1.602e-17 N \\
 a &= 1.602e-17 / m_p = 1.602e-17 / 1.673e-27 = 9.575e9 m/s^2 \\
 a_E M &= 9.575e9 \times 11 \times 1e-12 = 1.053e-1 m/s^2
 \end{aligned}$$

Step 5: Time-Reversal Correction

$$1 + f_{TRZ} = 1.1$$

Step 6: Final Solution

$$\begin{aligned} g_{\text{Mice}} &= (2.950e - 10) \times (1.02169) \times (0.7362) \times (1.1) + 1.053e - 1 \\ &= 2.950e - 10 \times 1.02169 = 3.014e - 10 \\ &\times 0.7362 = 2.219e - 10 \\ &\times 1.1 = 2.441e - 10 \\ &= 2.441e - 10 + 1.053e - 1 \\ &\approx 1.053e - 1 \text{ m/s}^2 \end{aligned}$$

4. Physical Interpretation

The Mice Galaxies result $(1.053 \times 10^{-1} \text{ m/s}^2)$ matches the Antennae galaxy result confirming UQFF's prediction that induced magnetic field enhancement is the universal discriminator for colliding galaxies. The elevated $B = 10 - 4T$ (100 times isolated spirals) drives the Aether correction from $\sim 10^{-2}$ to $\sim 10^{-1} \text{ m/s}^2$. Classical gravity $(2.950 \times 10^{-10} \text{ m/s}^2)$ is negligible. The merger factor $M_{\text{merge}} = 0.2638$ reflects $26 \tau_{\text{merge}} = 400 \text{ Myr}$. The Hubble observation confirms this is an active starburst system, validating the enhanced B-field assumption.

5. UQFF Framework Advancement

- Dual-galaxy merger formalism established: $M_{\text{merge}}(t)$ with $\tau_{\text{merge}} = 400 \text{ Myr}$
 - Starburst B-field enhancement (10^{-4} T) is the UQFF signature of merging pairs
 - Confirms UQFF converges to same result for similar merger stages (Mice $\approx$ Antennae)
 - Merger timescale $\tau = 400 \text{ Myr}$ establishes the UQFF galaxy-collision constant
-

6. Conclusions

The Master UQFF gravity equation for NGC 4676 (Mice Galaxies) yields $g_{\text{Mice}} \approx 1.053 \times 10^{-1} \text{ m/s}^2$, dominated by the starburst-enhanced Aether electromagnetic correction ($B = 10 - 4T$). The merger mass-loss term $M_{\text{merge}} = 0.2638$ reflects early post-encounter dynamics. The result agrees with the Antennae system, establishing 1 m/s^2 as the universal UQFF scaling for major starburst mergers. The Mice Galaxies join the Antennae as canonical UQFF merger benchmarks, validating the B-field enhancement model for early-stage colliding galaxies.

PAPER_769, CP4 class #353. v5.40.

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

Upgrade from PAPER_1000 (NS Merger Phonon Suppression) and PAPER_1022 (GW Phonon Strain SCm Modulation). See also PAPER_1011-1012 for GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^{(3)}} \right)$$

where: - $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$ is the phonon modulation factor - $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$ is the SCm phonon resonance frequency - $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$ is the third-order Ramanujan summation - Θ is the Heaviside step ensuring $H_{\text{SCm}} \geq 0.5$ (phase-transition threshold)

Physical mechanism: The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy ΔE_{BCS} of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at $M \approx 2.5 M_{\odot}$.

Calibration (canonical): $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$, $\beta_i = 0.603$, $H_{\text{SCm}} \approx 0.99$.

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_Bi_i\}}$ jet modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ kg/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile), PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044 (Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c} \right)^2 \right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad (\text{vs standard } b = 0.20)$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\text{--}4\%$ at cluster cores, partially resolving the Planck SZ–CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^{(3)} \cdot \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.p`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(- \exp! \left(- \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.158 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 71, \quad n_{\text{channel}} = 16/26$$

Since $p_{\text{DVP}} = 71$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos! \left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.158	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 71$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1 \dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ — $2(UQFF \text{ vacuum term})$	Planck 2018 $1.114 \times 10^{-52} \text{ m}^{-2}$	PASS Consistent	
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	PASS Consistent
UQFF buoyancy signature	$F_{\text{U_Bi_i}}$ unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{\text{U_Bi_i}}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1000	NS Merger $F_{\text{U_Bi}}$ Strain Suppression & BCS Gap
PAPER_1001	SMBH Binary Merger $F_{\text{U_Bi}}$ Phonon Damping
PAPER_1011	GW170817 NS Merger $F_{\text{U_Bi_i}}$ 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded $F_{\text{U_Bi_i}}$ with S26(3)
PAPER_1014	SMBH Merger Inspiral-Coalescence-Ringdown
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN $F_{\text{U_Bi_i}}$ Jet Modulation
PAPER_1010	TON618 AGN $F_{\text{U_Bi_i}}$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1039	SCm Galaxy Cluster Buoyancy Profile ICM Beta-Model
PAPER_1040	SCm Cluster Merger Shock Mach Number Phonon Damping
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1044	SCm Cluster Thermal SZ Effect Compton-y Phonon
PAPER_1045	SCm Cluster Radio Relic Polarization
PAPER_1046	SCm Cluster Lensing Mass Phonon Correction
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1035	Kilonova Buoyancy Light Curve r-Process

18 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\ramanujan\appendices}.py`. For detailed derivations, see `PAPER_840/851/852/855`.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_{s26_coupling}.py</code>	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_{scm_cross_section}.py</code>	Sym-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor (1+[SSq]*n/26)
<code>kozima_{wstp_kernel}.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \times \sigma_n^{\text{SCm}}(\omega) \times \Phi_{\text{phonon}} \times (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \times \exp[-(\omega - \omega_{\text{SCm}})^{2/(2 \times \text{Gamma}2)}] \times (1 + [\text{SSq}] \times n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_{polylog_s26}.py</code>	$Li_{26}([\text{SSq}])$ via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_{wstp_kernel}.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} \times Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_{theta_q26}.py</code>	$f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series	Proper q-Pochhammer (a;q)_n

Core equations: - $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q;q)_n^2$ - $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^{2;q}2)_n$ - $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
<code>ramanujan_{pi_uqff}.py</code>	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
<code>mock_{theta_pi_wstp_kernel}.py</code>	11-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n \times (1103+26390n) \times W_{26}(n) / C_{26}$ where $W_{26}(n) = \text{Prod}_{\{i=1\}^{\{26\}}} [1 + [\text{SSq}] \exp(-kappa_i \times n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in *CondensedPhysics.py*, *CondensedPhysics2.py*, *MAIN_{1_CoAnQi}.cpp*, and Wolfram kernels (*uqff_{kozima_kernel}.wl*, *uqff_{s26_kernel}.wl*, *uqff_{mock_theta_pi_kernel}.wl*).

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